What makes up a color?

The color is made up of three attributes: Hue, Saturation, and Value.

Hue (H)

Defines the spectrum.

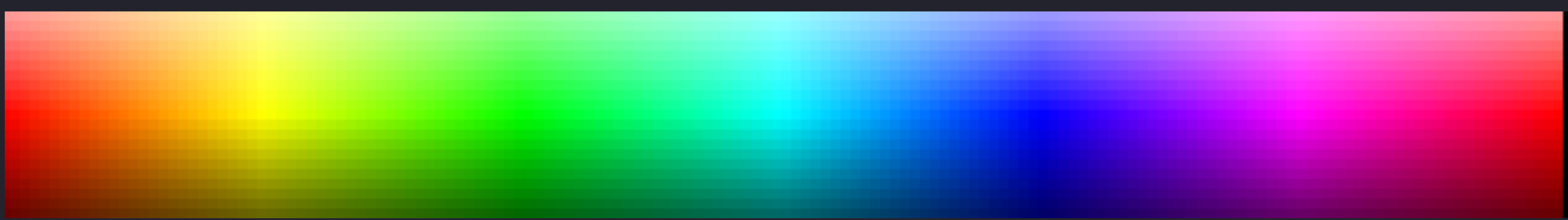
Saturation (S)

Defines intensity of a spectrum.

Value (V)

Defines the tone of a spectrum.

Spectrums



How do items get colors?

When all color spectrums come together, we see white.

When there are no color spectrums, we see black.

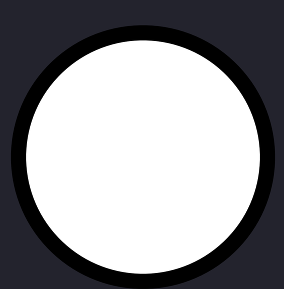
Light with all spectrums shines on an object.

The object absorbs some spectrums and reflect others.

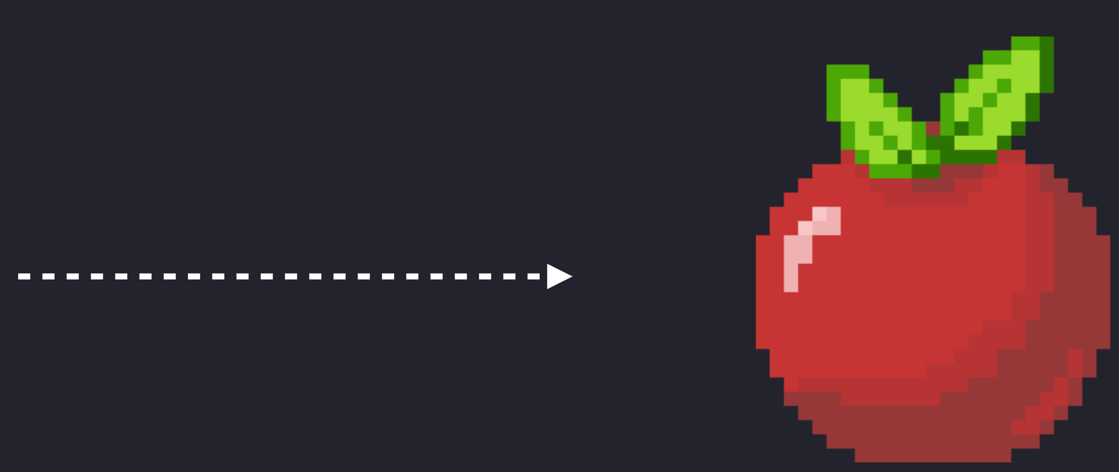
The reflected spectrums define the colors of an object.

Black = No Spectrums

White = All Spectrums

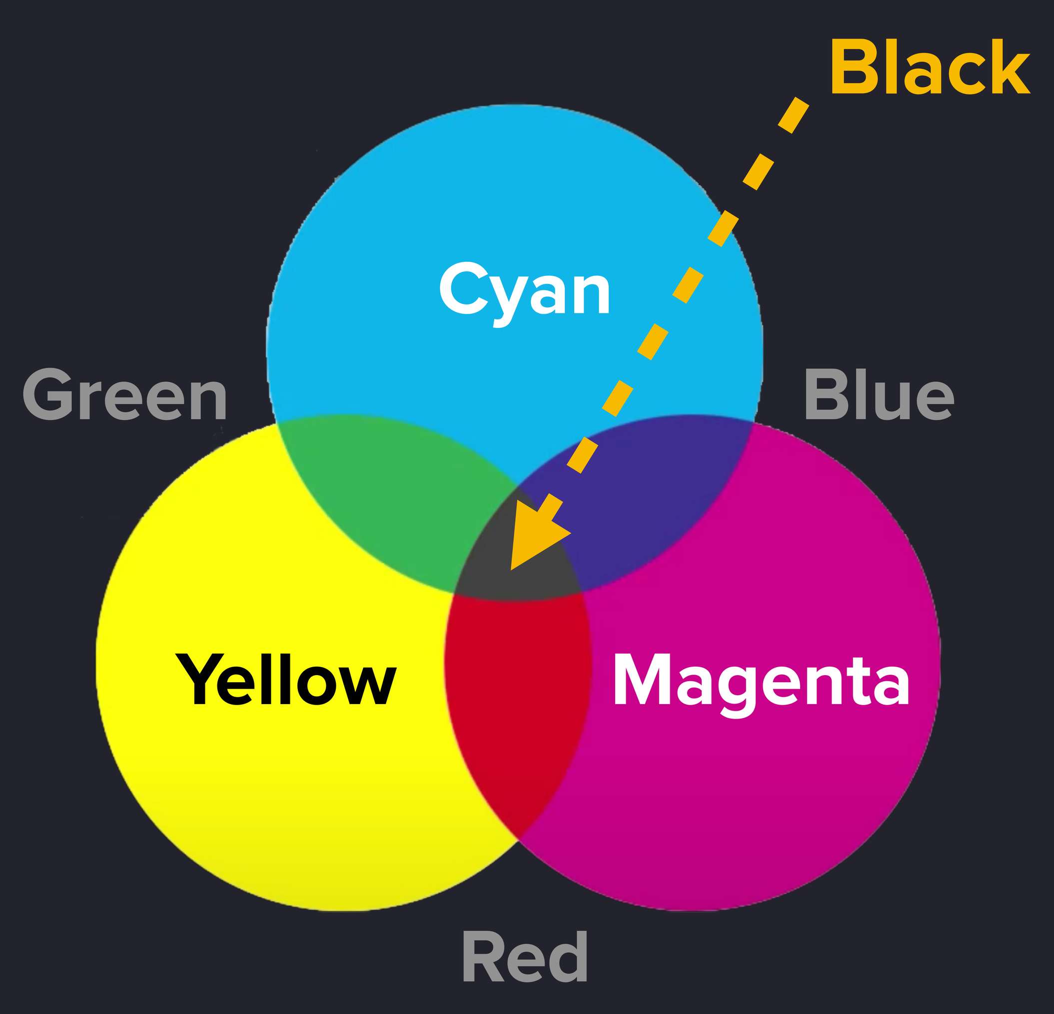


Light with all Spectrums



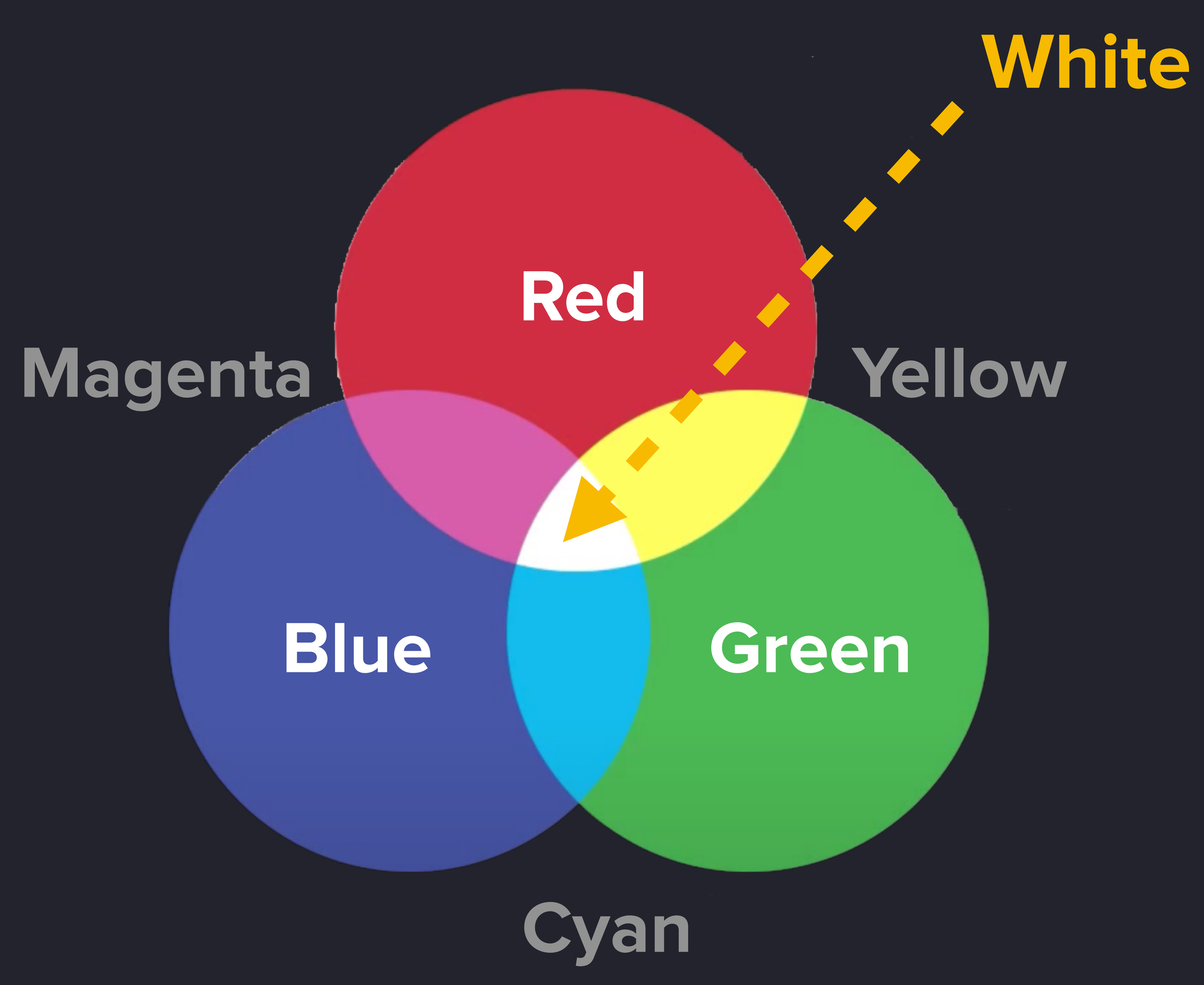
Apple absorbs all spectrums except for red. Hence the apple is red.

Color Models



CMYK

Subtracts colors
Four channels: Cyan, Magenta, Yellow, Black
Used in printing



RGB

Additive colors
Three channels: Red, Green, Blue
Used in displays

Spot

Individual colors and shades
Used in offset printing (posters, billboards, stationary, etc)

Color Wheels

Why color wheels are important?

They help you design your color palette in the way that your Hues, Saturation and Values are balanced overall.


Monochrome
One color + shades


Adjacent
One color + 2 adjacent colors


Triad
Three colors


Tetrad
Four colors

Designing your color palette

Start with a limited palette.
Add up as you go.

Use color wheels
to keep Hues, Saturation and Values consistent.

Do not use Opacity.
Shift Hues, Saturation and Values instead.

For print purpose, use CMYK model, or make sure Spot model is available.